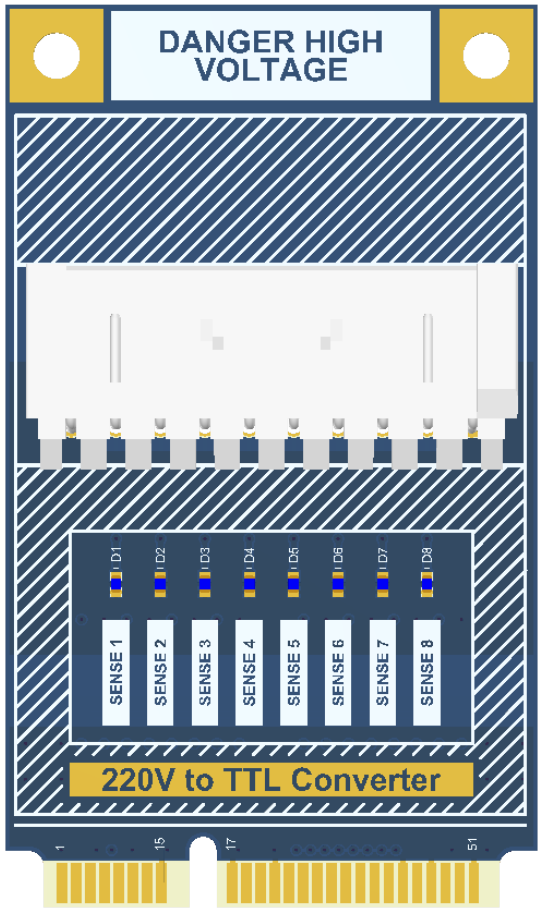
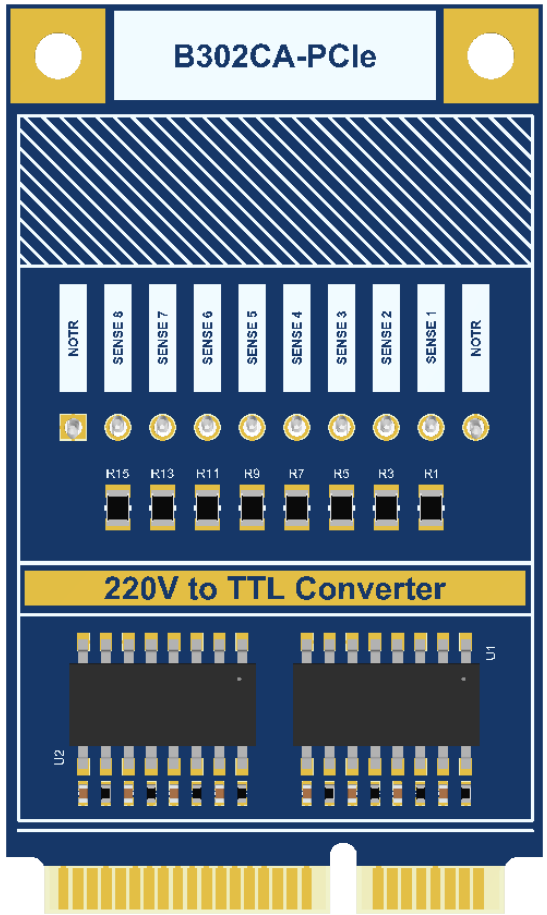
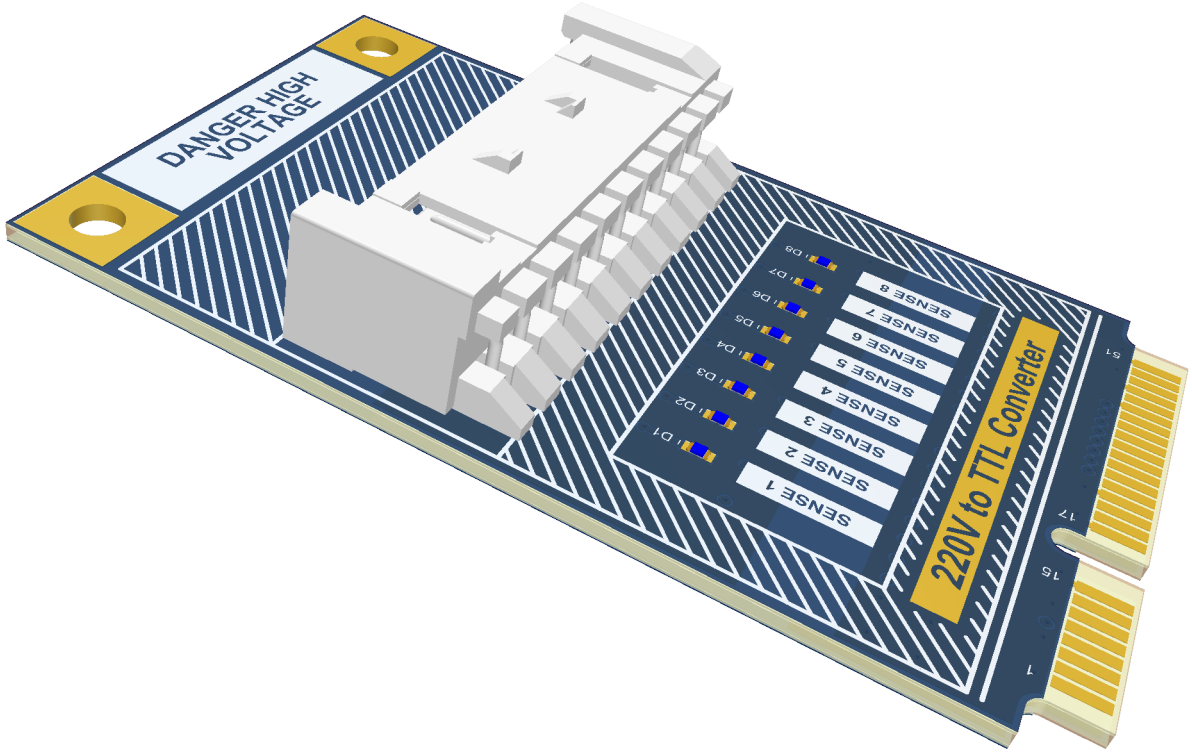
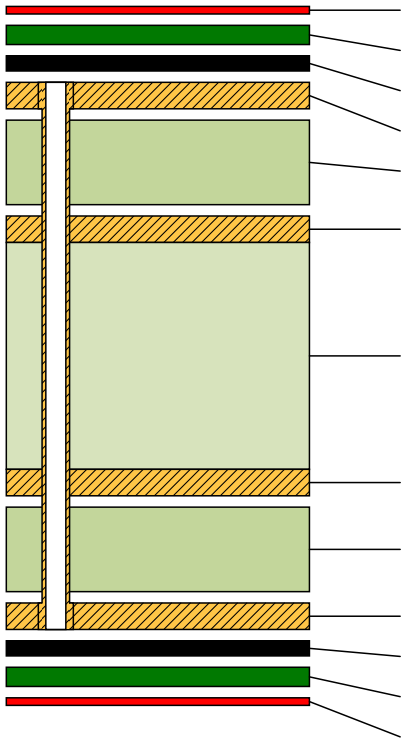

DANGER HIGH VOLTAGE
D1
D2
D3
D4
D5
D6
D7
D8
SENSE 1
SENSE 2
SENSE 3
SENSE 4
SENSE 5
SENSE 6
SENSE 7
SENSE 8
220V to TTL Converter
1
15
17
51


B302CA-PCIe
NOTR
SENSE 8
SENSE 7
SENSE 6
SENSE 5
SENSE 4
SENSE 3
SENSE 2
SENSE 1
NOTR
R15
R13
R11
R9
R7
R5
R3
R1
220V to TTL Converter
U1
U2


DANGER HIGH VOLTAGE
220V to TTL Converter
SENSE 1
SENSE 2
SENSE 3
SENSE 4
SENSE 5
SENSE 6
SENSE 7
SENSE 8
D1
D2
D3
D4
D5
D6
D7
D8
1
15
17
54

Layer Stack Legend

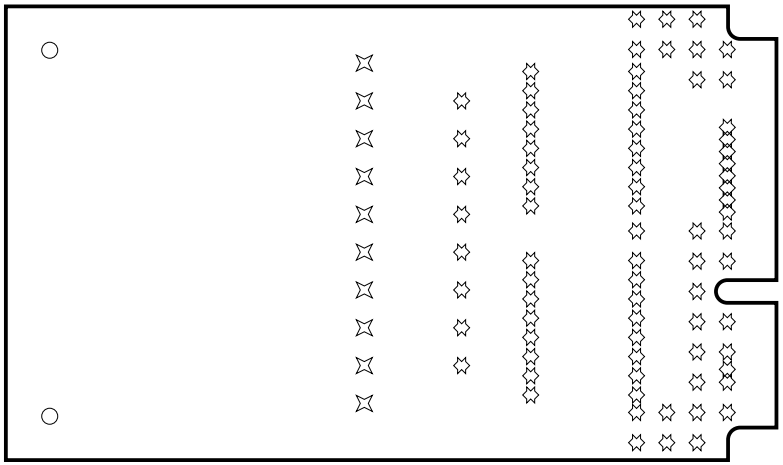


| Material | Layer | Thickness | Dielectric Material | Type | Gerber |
|---|---|---|---|---|---|
| | Top Overlay | | | Legend | GTO |
| Surface Material | Top Solder | 0.03mm | SM-001 | Solder Mask | GTS |
| Nickel, Gold | Top Surface Finish | 0.02mm | | Surface Finish | |
| **CF-004** | **Top Layer** | **0.04mm** | | **Signal** | **GTL** |
| *Prepreg* | | *0.11mm* | *PP-015* | *Dielectric* | |
| **CF-004** | **Inner Layer 1** | **0.04mm** | | **Signal** | **G1** |
| *Core* | | *0.61mm* | *Core-038* | *Dielectric* | |
| **CF-004** | **Inner Layer 2** | **0.04mm** | | **Signal** | **G2** |
| *Prepreg* | | *0.11mm* | *PP-015* | *Dielectric* | |
| **CF-004** | **Bottom Layer** | **0.04mm** | | **Signal** | **GBL** |
| Nickel, Gold | Bottom Surface Finish | 0.02mm | | Surface Finish | |
| Surface Material | Bottom Solder | 0.03mm | SM-001 | Solder Mask | GBS |
| | Bottom Overlay | | | Legend | GBO |

Total thickness: 1.06mm

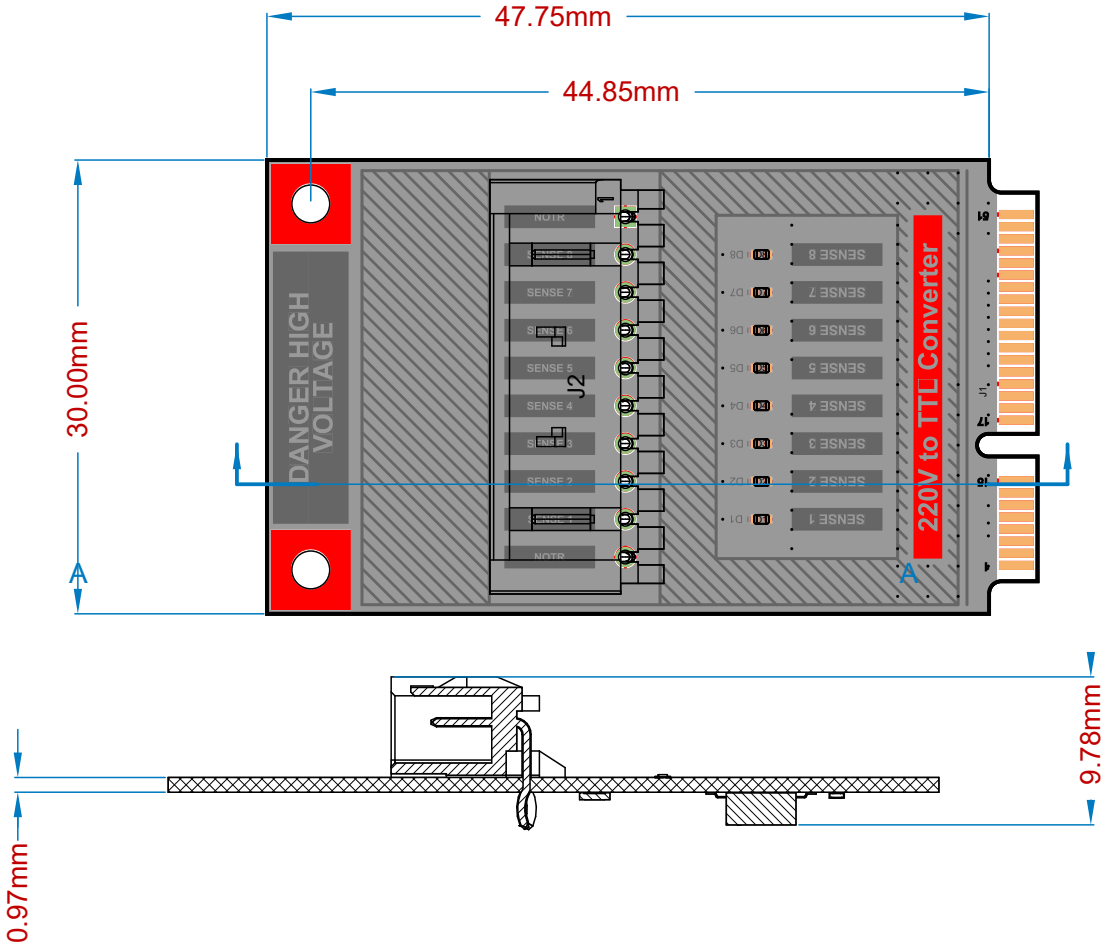
Bill Of Materials

| Designator | Quantity | Manufacturer Name | Manufacturer Part Number |
|---|---|---|---|
| C1, C2, C3, C4, C5, C6, C7, C8 | 8 | KEMET | C0402C475M9PACTU |
| D1, D2, D3, D4, D5, D6, D7, D8 | 8 | ROHM Semiconductor | SML-P12MTT86R |
| J2 | 1 | Molex | 53426-1010 |
| R1, R3, R5, R7, R9, R11, R13, R15 | 8 | Vishay Dale | CRCW0805220KJNEAHP |
| R2, R4, R6, R8, R10, R12, R14, R16 | 8 | Vishay | CRCW040247K0FKED |
| U1, U2 | 2 | Vishay Semiconductors | TCMT4100 |



Drill Table

| Symbol | Count | Hole Size | Plated | Hole Tolerance |
|---|---|---|---|---|
| ✩ | 77 | 0.20mm | Plated | |
| ⌘ | 10 | 1.00mm | Plated | |
| ○ | 2 | 2.60mm | Plated | +/-0.05mm |
| | 89 Total | | | |


47.75mm
44.85mm
30.00mm
A
DANGER HIGH VOLTAGE
J2
SENSE 7
SENSE 5
SENSE 4
SENSE 2
NOTR
SENSE 8
SENSE 7
SENSE 6
SENSE 5
SENSE 4
SENSE 3
SENSE 2
SENSE 1
220V to TTL Converter
A
9.78mm
0.97mm